

CHAPTER 1

Blockchain architecture, taxonomy, challenges, and applications

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1. Introduction to blockchain

Blockchain is a digitized, distributed ledger for records. It is simply a distributed database recording transactions in chronological order. To tamper with a blockchain is almost impossible. As the name blockchain indicates, a blockchain is nothing but a chain of blocks that stores data. A blockchain stores information in a different way than a typical database. Whenever data should be added, a block is newly created; new data is stored in the block, and then the newly created block will be chained to the previous block. The block creation is done chronologically and in a linear fashion.

There is no particular type of information that should be stored in a blockchain, but the most common application so far for blockchain has been as a record for transactions. The working steps of a blockchain are shown in Fig. 1.1. In blockchain, there is not a central server or third party that has complete control over the network. Blockchain uses a peer-to-peer network to group its computers. This means all the information that is stored in a blockchain is accessible to all the computers present in the network. So, if one person tries to tamper with the data in the blockchain, all others present in the blockchain can cross-reference one other, and that one copy of the blockchain will stand out as the culprit (Wüst and Gervais, 2018).

It is not necessarily impossible to change data in a blockchain. If the majority (i.e., 51%) of the network claims the change of data, then the data can be modified. As the network grows, it is practically impossible to tamper with the data. Even if someone wants to venture on this near impossible mission, the cost of resources required to tamper with data in a

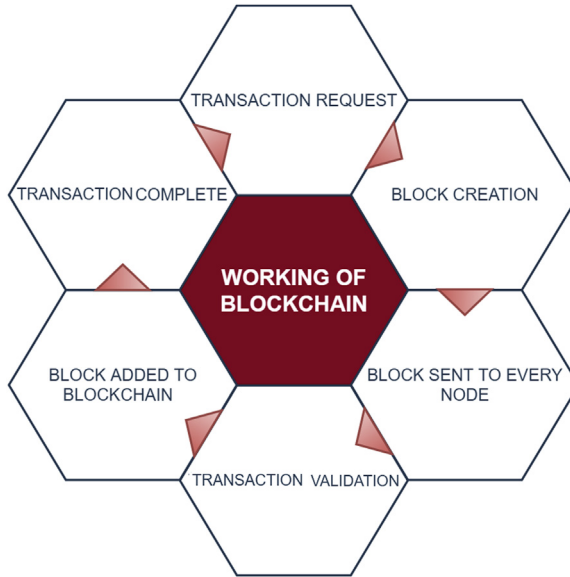


Figure 1.1 The working of blockchain.

blockchain is much more than the outcome. This property where the control is with no one member of the network is called decentralization. It is also possible to create a blockchain that has a centralized architecture.

Blockchain has several applications other than storing transaction data like in cryptocurrencies. Wherever there is a requirement for security, decentralization, or privacy, we can incorporate blockchain technology into that application. Some blockchain applications are in supply chain management, the energy sector, and the business sector. And in the government sector, voting and identity management are two of the applications. In [Section 7](#) a detailed explanation of various blockchain applications are given.

1.1 Motivation

We live in a world where the technology advances every day. Every once in a while, there emerges a technology that completely changes the way that we perceive the world. Generally, a technology is born if there is a need to solve a problem that is not being solved using existing technologies. One such technology that is very much popular today is blockchain. Blockchain is simply a database in which the information is stored as blocks of data. Trust is the basis of any financial transaction. Now, to do any type of financial transaction, we rely on intermediaries such as banks. The trust gap is achieved with the help of intermediaries, and the intermediary

demands a fee for their services. But by using blockchain for financial transactions, one can achieve a far greater level of trust and achieve this trust with a fraction of what one pay for intermediaries.

1.2 Blockchain analogy

Let us visualize a huge treasury in a bank. The treasury vault is filled with rows of deposit boxes. Each deposit box is made of glass, allowing everyone to visualize the contents of the deposit boxes, but people only have access to their own box. When a person opens a new deposit box, he or she receives a key that is unique to that box. This is the fundamental concept of cryptocurrencies based on blockchain. Anyone can see the contents of all other addresses.

1.3 Characteristics of blockchain

Each block is built on top of the previous block and uses the block's hash to form a chain. Validating and confirming blocks over the chain is handled by miners. Blocks created are cryptographically sealed over the blockchain, which means that it is nearly impossible to delete and modify data over the blockchain (Xu et al., 2019). Consensus algorithms make sure that all transactions are validated and only added once over the blockchain. The miner receives a reward for running the consensus algorithms; the current reward is 12.5 BTC in the Bitcoin blockchain and 2 ETH in the Ethereum blockchain. All the blocks added are in chronological order and time stamped.

1.4 Summarizing blockchain

A blockchain is a digitized, distributed, consensus-based secure storage of information protected from revision and tampering over the peer-to-peer network. Blockchain is a digitized store for information in the form of transactions. It is distributed. Thus, nobody controls it. Consensus algorithms ensure security and immutability. When new data arrive, they are stored in a block and chained to the blockchain after validation using a consensus mechanism. Data gets recorded in chronological order. Everyone presents over the network can view the transactions. The main elements of blockchain are privacy, security, and efficiency (Fig. 1.2).

1.5 Bitcoin and blockchain

A blockchain is a distributed database of records. Every smart contract or transaction in the public blockchain is confirmed by any one of the

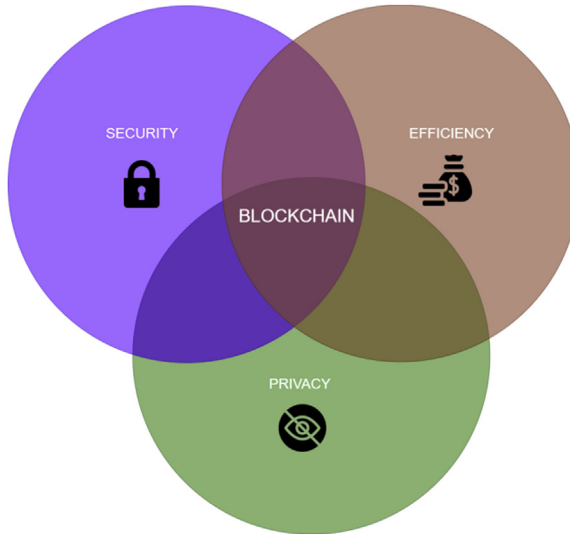


Figure 1.2 Key elements of blockchain.

consensus mechanisms. Transactions are encrypted and cannot be replicated or altered. Currently, the most famous blockchain application is Bitcoin. Blockchain is able to transfer anything from stocks to property rights and cryptocurrency with ease and without the need to go through a third party.

Bitcoin is a digital cryptocurrency, created and withheld digitally on your computer or in a wallet that is virtual. It is distributed, i.e., not centralized, so there are no single entities like persons, organizations, or banks that control the currency. It was started in 2009 to get rid of third-party payment processing intermediaries. The blockchain is the supporting technology that upholds the Bitcoin transaction ledger.

1.6 Wallets, digital signatures, and protocols

A blockchain wallet is similar to a digital wallet in that it allows participants to manage their cryptocurrency. A wallet lets users generate a private key and public address. The private key is used to send the transaction, and a public address is used to receive the transaction. No observable archives of identity about who did what deal or contract with whom are present; only the address of a wallet is visible in the transactions. Different types of blockchain wallets include paper wallets, web wallets, mobile wallets, desktop wallets, etc. Digital signatures are similar to real-world paper signatures as a means to verify a person's identity. Digital signatures use cryptography, which is more secure than handwritten signatures (Xu et al., 2017).

A private key is used to sign messages digitally. The recipient can verify it with the sender's public key. Every transaction that is executed on the blockchain is digitally signed by the sender using their private key. SSL is an example of a digital signature.

1.7 Main contributions of this chapter

The main contributions of this chapter are the following:

- (1) The real-life applications of blockchain technology are explained in detail in the latter part of the chapter. Ten applications are listed.
- (2) This chapter explains the challenges of blockchain technology like industry challenges and types of blockchain attacks.

1.8 Organization of the chapter

The first section of this chapter gave a basic understanding of blockchain. [Section 2](#) contains the brief history of blockchain. [Section 3](#) describes the advantages of using blockchain over traditional technologies. This section also explores the key benefits of using blockchain such as decentralized control and confidentiality. [Section 4](#) explores the architecture underlying blockchain and various consensus mechanisms like proof of work. This section also discusses the types of blockchain and blockchain ecosystem. Later in this section, various popular blockchains are listed. [Section 5](#) gives the overview of blockchain evolution in fields other than cryptocurrencies like Bitcoin. [Section 6](#) explains the various challenges of blockchain technology, mainly industrial challenges, and attacks against blockchain. [Section 7](#) explores the applications of blockchain in detail. In this section, 10 real-world applications of blockchain are discussed in detail. [Section 8](#) concludes with an assessment of future work.

2. History of blockchain

In early 1990s, the blockchain technology was described with the intent to avoid tampering and time stamp digital data. Almost 20 years later, this technology was adapted by Satoshi Nakamoto for the creation of a cryptocurrency known as Bitcoin. The first practical use case of blockchain technology was Bitcoin.

2.1 Use of blockchain in Bitcoin

Satoshi Nakamoto published a research paper in 2008 entitled “Bitcoin: A peer to peer electronic cash system” that stated that the transactions of this

cryptocurrency Bitcoin could take place without the involvement of a third party. With the arrival of Bitcoin in 2008 the blockchain technology came to light even though it had been discovered in the early 1990s. A new protocol was released after a few months that began the concept of a genesis block with 50 coins. This new protocol was an open-source program that gradually became a part of Bitcoin network.

2.2 Ethereum and smart contracts

In 2013, Canadian-Russian programmer Vitalik Buterin, who initially contributed a significant amount of code to Bitcoin codebases, went on to build the cryptocurrency popularly known as Ethereum. He saw many limitations in Bitcoin technology, so he built another cryptocurrency. Unlike Bitcoin, which just records transactions of Bitcoin cryptocurrency, Ethereum helps its users to record digital data of their cars, properties, yachts, contracts, etc.

The launch of Ethereum took place in 2015, and it has the functionality of smart contracts. Based on a set of criteria established in the blockchain, with the use of smart contracts, we can perform logical operations automatically (Syed et al., 2019). For example, a smart contract can be formed to place a bet between two parties. Then, both parties would send cryptocurrency, which is their part of the bet, and upload the contract to the Ethereum blockchain. Each of their cryptocurrencies is held by the software underlying Ethereum blockchain, and after the bet ends, the smart contract would check who is the winner using logical operations and then post the earnings to the winner.

3. Benefits over traditional technologies

Blockchain technology was discovered due to the roadblocks of existing technologies. With the use of blockchain, there exists a decentralized control, in contrast to that of traditional technologies, which support centralized control. The usage of blockchain technology ensures transparency and integrity of its users. The advantages and disadvantages (Fig. 1.3) are discussed later in the chapter in detail. There are a few benefits of blockchain over traditional technology: enhanced security to the user, confidentiality, faster processing, etc.

3.1 Decentralized control

Blockchain technology supports multiple parties that have no trust in each other to share data without the need of a centralized control. It removes the

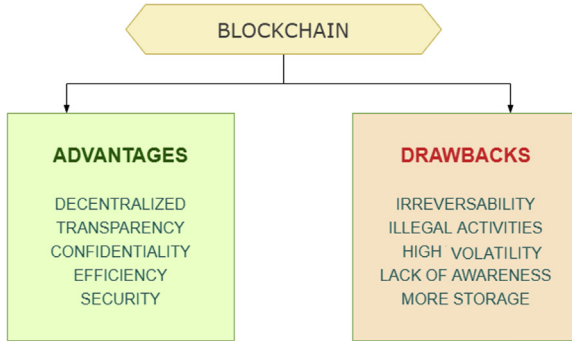


Figure 1.3 Advantages and drawbacks of blockchain.

dangers that are caused by centralized control. With a centralized control over information, anyone with adequate access to the organization can tamper with or corrupt the data within. Cost savings are also provided; usually billions of dollars are spent on safeguarding central repositories from hackers. Blockchain provides a similar public shared system of record concurrently for everybody who is connected to the network. The trust is established by the cryptographic protocols running behind the blockchain technology. All the parties must agree to make a change in blockchain, which is nearly impossible.

3.2 Integrity and transparency

Blockchain technology replaces traditional data storage technology as it is publicly verifiable, aided by transparency and integrity. A user present in a blockchain network can trust that the information they are accessing through a blockchain is tamper-proof and uncorrupted from the instant it was logged and recorded. Every user can authenticate the information added over the blockchain. Blockchain does not leave behind past records, and as the data is added to the blockchain, it appends and grows while also providing current information (Zheng et al., 2018). Merkle trees ensure the integrity of the data by hashing the transactions to a single root.

3.3 Confidentiality

The blockchain is an openly distributed ledger. Yet a private system can be established to maintain confidentiality. Data confidentiality in blockchain guarantee that single entities like persons or organizations that are prohibited from getting information are not approved obtain it. Permissioned blockchains have arisen as a substitute to public blockchain to meet the

needs of the industry for having recognized and distinguishable members. Solutions like Hyperledger Fabric blockchain and Block Stream offer rich sets of permissions to maintain confidentiality in the system.

3.4 Faster processing

The traditional banking process takes days to settle, but the blockchain has reduced that time to nearly 2 min or even seconds. Everybody has the access to the same data, and it is easy to build trust among one another without the necessity for any third parties or intermediaries. Moreover, tracking of products could also be made efficient by uploading the data into a blockchain. Digital assets and the trustless system make sure that the data is protected and transacted efficiently.

3.5 Enhanced security

The transactions are encrypted and linked to the previous transactions. Information is stored across a network of computers instead of a single server. Blockchain prevents fraud and unauthorized activity. Cryptography protocols make sure that the data is thoroughly secure. Safeguards are in place from denial of service (DoS) attacks as the data are present on all the nodes connected to the network. The cryptographic fingerprint (hash of the block) is unique for each block.

4. Blockchain architecture

A blockchain holds a chain of blocks in which any type of information can be stored, like in a database. The information that is stored in the blocks and chained together is secure because this information is available to any computer that is present in that network. The computers in this network are grouped together in a peer-to-peer (P2P) architecture, as shown in [Fig. 1.4](#). In simple words, blockchain is a group of computers linked together with no central server. Blockchain is decentralized because the network of computers present in a blockchain has no central server.

4.1 Understanding architecture

Every block in a blockchain has a block header. The block header is useful to differentiate each block in a blockchain network. It contains the time-stamp, version, and Merkle root of the block. One of the common algorithms that checks the legitimacy of a block is proof of work (PoW). To make the PoW operate requires two entities or fields: the difficulty target

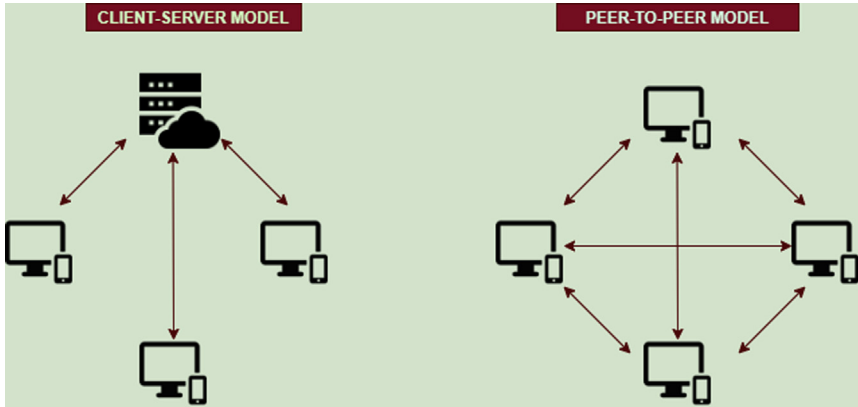


Figure 1.4 P2P and client-server model.

and the nonce. So, the block header should have the PoW fields of difficulty target and nonce. The last field that the block header should include is the previous hash. The blockchain's main property is its immutable nature. This immutable property is made possible by including the previous block hash in the calculation of the present block hash.

By doing so, if anyone wants to change a block in the blockchain, they are forced to change all the other blocks present in the blockchain. Even after replacing all the blocks, one cannot achieve the results that are expected to follow. This feat would be successful if the replacing of blocks took place for at least 51% of nodes present in the network. It is impossible in the case of blockchain as its network is vast, so to change the data in at least 51% of nodes is insurmountable.

4.2 Consensus mechanisms

Blockchain are decentralized systems that contain diverse members who act on condition of inducements they obtain and the data that is accessible to them. When a fresh transaction gets introduced on the network, computers connected to the network have the choice to either take in that transaction to their replica of ledger or to overlook and ignore it. When the 51% or more of the computers that comprise the network decide on a solitary state, the consensus is attained.

Blockchain is not controlled by a central authority, so it has decentralized ledgers. Due to the digital information kept and maintained in these ledgers, bad actors have significant monetary inducements to attempt to cause problems (Lin and Liao, 2017). PoW is a solution that involves

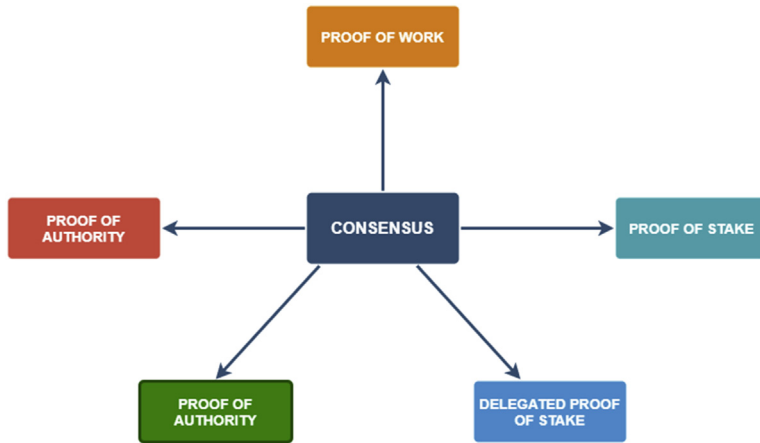


Figure 1.5 Types of consensus mechanisms.

probability, known as the Byzantine Generals problem (Lamport et al., 1982). It follows the longest chain rule where miners shift to the chain that is being more worked upon. When a miner cracks the puzzle and authorizes the block, all the computers present in the network will confirm if the block is binding or valid and append it to their replica of the blockchain. The computers in advance have a necessity to attain a consensus on the validity; only then will the blockchain network synchronize, and the state of the chain of blockchain will update. Various types of consensus mechanisms are shown in Fig. 1.5.

4.3 Various consensus mechanisms

4.3.1 Proof of work

PoW is a type of consensus algorithm where the miners contend to resolve a tough mathematical problem built on a cryptographic algorithm involving hash. This proof shows that a miner spends a lot of resources and time to resolve the difficult mathematical problem. When a block is resolved, the transaction data contained are thereby considered valid and stored in the blockchain.

By mathematical problem, we mean a function involving a hash. In the hash function the miners solve a puzzle to discover the input for which the output is already known. For example, consider integer factorization, which involves in what way to write a number as the product of two numbers, and the guided tour puzzle protocol, which involves the requirement of the calculation of functions that involve hash, if the server is

suspected of a DoS attack for some computers in a predefined order (Halaburda, 2018). Miners collect a payment once they crack the difficult mathematical problem, which involves hash functions. For example, in Bitcoin, miners receive 12.5 Bitcoin for solving the puzzle. Miners can also receive transaction fees in addition to rewards.

4.3.2 Proof of stake

Proof of stake (PoS) is an alternative algorithm to authenticate transactions and attain the consensus of a distributed ledger. PoW algorithm pays miners who crack difficult mathematical problems with the final goal of confirming transactions and generating fresh blocks. On the other hand, in the PoS algorithm, the maker of a fresh block is selected in a deterministic way, which depends on their affluence and position in the blockchain (Tapscott and Tapscott, 2017). At the start of the chain, all the digital currencies are created. Their number never changes. Here, in PoS, the miners only take the payment of transaction fees. So, the miners present in the PoS system miners are called forgers.

4.3.3 Delegated proof of stake

People in a specific blockchain network vote for spectators to safeguard their computer network. Let us imagine a reward system where only the top 50 spectators are rewarded for their service, and only the top 10 make a consistent salary. As it creates healthy competition, many want to become a spectator, thus providing hundreds of backup spectators. The strength of the vote of a person is resolved by number of tokens they hold. People who have more tokens will impact the blockchain network more than those who have fewer tokens. If a spectator starts acting like a schmuck or halts undertaking work in safeguarding the blockchain network, persons in the blockchain network can confiscate their votes, which basically means the lousy actor is thrown out of the blockchain community.

Voting is continuously taking place in a delegated PoS consensus mechanism. Delegates are elected as spectators. A delegate becomes a cosigner on an individual account that has access to propose certain changes that are required for the parameters of the network. The genesis account is the name of this account. These network restrictions comprise the whole lot from block sizes to transaction fees, spectator pay, and lump intervals.

4.3.4 Proof of authority

The proof of authority (PoA) consensus mechanism is fundamentally an improved PoS consensus that controls identity as the system of stake rather

than token staking. The group of validators is typically supposed to stay relatively small to ensure manageable security and efficiency of the network. Individuals under PoA earn the right to become a validator, which is why there is no incentive to retain the position they hold (Tasatanattakool and Techapanupreeda, 2018). Validators are required to formally verify identity either on the chain or some public domain. The eligibility to become a validator is difficult to obtain, where the individuals need to go through many steps to become a validator.

4.3.5 Proof of weight

Proof of weight (PoW) is a broad consensus mechanism based on the Algorand algorithm, which in turn specifies a new protocol known as Byzantine agreement (BA). BA protocol is highly scalable and secure. The PoW consensus model runs a committee where participants keep on changing, and the committee achieves the consensus for the network. Every user over the network has a weight attached to them, which is determined by the amount of money they hold in their account.

4.4 Types of blockchain

4.4.1 Public blockchain

A public blockchain, as its name indicates, is the blockchain that is available to everyone; in other words, it is a form of blockchain which is for the people, by the people, and of the people. There is no one in charge of the network, and anyone can contribute to reading, writing, and auditing the blockchain. More complex rules are present for safeguarding it from malicious actors. This is the most used blockchain, as shown at the top in Fig. 1.6. All decisions are made using a complex consensus algorithm. Computationally, these blockchain are expensive to mine and commit a block over the network. Examples are Bitcoin and Ethereum (Mettler, 2016).

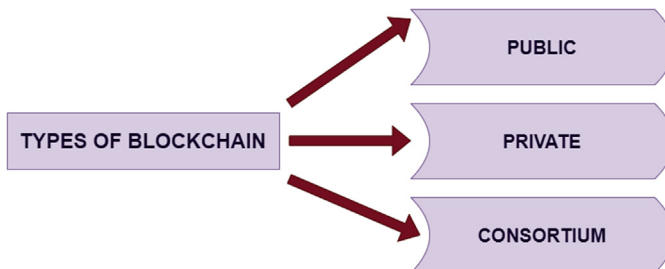


Figure 1.6 Types of blockchain.

4.4.2 Private blockchain

An individual or an organization privately operates a private blockchain, as its name suggests. Unlike public blockchain, there is an administrator or anchor that looks after essential things such as permissions and identities in a private blockchain. The consensus is attained on the notion of the dominant party who can provide mining rights to everyone or just keep those to themselves. Compared to a public blockchain, it is cheaper and faster since a person does not have to employ a huge amount of energy, money, and time to attain a consensus. It is less secure compared with the public blockchain.

4.4.3 Consortium blockchain

A consortium blockchain removes individual independence, which gets conferred in just one unit by means of a private blockchain. Here, in place of a single entity in charge, we have more than one entity in charge. A collection of organizations or representatives can make choices for the profit of the whole consortium blockchain network, as a way of attaining things more quickly and to have more than a single point of checks, which defends the whole blockchain ecosystem. In simple words, it is the best of both private and public blockchains. This blockchain type gives choices for access management and rights while controlling the same blockchain technology and earning its paybacks (Al-Jaroodi and Mohamed, 2019).

4.5 Blockchain technology stack

The stack is the base of all blockchain-based applications. It is a network of nodes spread across the world that runs algorithms to verify and commit transaction over the blockchain. This is a decentralized network and mostly based on open-source cryptography algorithms. This can be created as public, private, or consortium, and it can include certificate authorities and membership services for governments and enterprises.

4.5.1 Overlay networks

This is a network that achieves additional functionality without bootstrapping the original protocols. They benefit from the network effects and build on top of that. Some of the additional functionalities you achieve are smart contracts that involves in crypto agreements between two parties, storage for renting network and side chains which help to transfer tokens from one blockchain to a different blockchain.

4.5.2 User interface

Finally, a user interface is required for your users to connect to a blockchain (Scott et al., 2020). A user interface can be built with existing front-end mechanisms and can easily employ the power of blockchain by using APIs. Some platforms like Hyperledger also provide generators to produce a box user interface for the applications built upon them. Most blockchain platforms also provide connecting services to link applications to blockchain.

4.5.3 Decentralized protocols

This is the essential part of the blockchain technology stack as it makes sure the blockchain layer and overlay networks do not depend on a single entity for validation and transactions. It helps to create decentralized P2P datasets. Decentralized protocols can be based on asset mapping, social identities, verification algorithms, or communication channels.

4.5.4 Support system

For building blockchain applications, APIs are required to interact with the underlying technology (Subramanian, 2017). One can directly build a solution using the existing APIs, without depending and working on a ledger. Most APIs respond to JSON format, so can be easily integrated with existing front-end technologies. You can use open-source APIs like Bitcoin and Ethereum, or you can choose from commercial APIs such as Block.io or BlockCypher for more boxed-up functionalities.

4.6 Blockchain ecosystem

The blockchain ecosystem is currently running some major projects, and more are in the pipeline (Saber et al., 2019). Some of the major projects on blockchain are Bitcoin (which introduced blockchain to the world), Ethereum (which came with the concept of smart contracts where two parties adhere to certain rules and create a trust, opening the world for more decentralized applications), NEO (which positioned itself as the “Chinese Ethereum,” and it bought Python as the main language for the creation of applications), and Stellar (which is trying to make cross-border transactions simpler). Stellar comes with extensive APIs that help developers build applications fast, thus reducing the time to market for applications.

4.6.1 Blockchain miners

To make a blockchain functional and maintain its integrity, a large network of independent nodes around the world is needed to be continuously maintained. In private blockchain, a central organization has the authority

over every node on the network. In the case of public blockchain, on the other hand, anyone can set up their computer to act as a node. The owners of these computers are called miners (Guo and Liang, 2016). Since the integrity of the blockchain is directly related to the number of independent nodes on the network, there also needs to be some incentive to mining. Different blockchains utilize different mining systems, but most of them contain some form of an incentive system or a consensus algorithm.

4.6.2 Blockchain developers

Blockchain technology is built through the potential of developers working behind it. A strong team of developers can lead to a successful blockchain project. Currently, there are two types of developers in the blockchain ecosystem: blockchain developers and dApp developers. Blockchain developers build new blockchains with different levels of functionalities and consensus algorithms. dApp developers work with decentralized applications that can run on blockchain, thus providing a similar functionality like Google Play Store over the blockchain technology. The development of smart contracts over the blockchain has opened the possibility for developers to create extensive applications and use cases for industries.

4.6.3 Blockchain users

Blockchain users are average people, like you and me, who make use of the blockchain or cryptocurrency to achieve some results (Niranjana Murthy et al., 2019). They can also be investors who buy cryptocurrencies to sell later. For creating a blockchain user base the technology or cryptocurrency should have some utility related to the problem being tackled. For example, Bitcoin serves as a major utility for payment of goods and services. Currently, there are over 50,000 merchants registered with Bitcoin, including Microsoft, PayPal, and Subway. Bitcoin was the first mover in blockchain, and its high utility as a payment system ensures that a large part of its ecosystem is based upon users.

4.6.4 Blockchain exchanges

Every blockchain project has a robust ecosystem working under it, and it always includes a decentralized exchange. These are developed by the blockchain team or a community of developers. A typical exchange is designed to find the cheapest rates of exchange between any two cryptocurrencies, making it more affordable to trade tokens and cryptocurrencies. Exchanges used for trading also might integrate with hardware wallets, or users can create their own wallet on the exchange website.

4.7 Various top blockchains in the contemporary world

4.7.1 Bitcoin

Bitcoin is a digital payment system and one of the very popular cryptocurrencies around the globe. It was founded in early 2010 and released as open-source software. Bitcoin is not operated by a single person or an organization; instead, it is operated by everyone in the network of the Bitcoin blockchain, which is typically a P2P network. Bitcoin gave us the first glimpse of the blockchain, and it is the first decentralized digital currency whose ledger is maintained by blockchain.

Bitcoin transactions take place directly between users, which does not include an intermediate or third party. In this P2P network, nodes verify the transactions that are done by users, and these transactions are recorded in a distributed ledger, which is available to all the nodes in the Bitcoin network. Bitcoin stands out from fiat modes of payment. Some of the features that make the Bitcoin different from fiat currencies are its simple setup process, transparency, anonymity, faster transaction speed, decentralized mechanism, and lower transaction fee (Zhou et al., 2020). The main problem with banks is that they make you go through a complex process to just create an account with them. But in the case of Bitcoin the configuration process is very straightforward, and moreover, it is free of cost. With banks, there is the process called “know your customer” in which you have to show your personal identification to create an account with that bank. But a Bitcoin user does not need to link any personal identification information to their Bitcoin address. In fact, the user can have as many Bitcoin addresses as they wish.

Every Bitcoin transaction is recorded in a linear, public, distributed ledger, in simple terms, a blockchain. Due to this immutable way of storing the transaction information, the Bitcoin once transferred to a Bitcoin address cannot be refunded. As there is no third party or any sort of intermediary involved in a transaction of Bitcoin, the payment process is much faster than any traditional banking system.

However, Bitcoin charges a small fee for international transfers because of the validity mechanism involved in the transaction processes (Cocco et al., 2017). A large chunk of this fee is claimed by the Bitcoin miners who validate the transaction and store the transaction in a block, which then is chained to the Bitcoin blockchain. The Bitcoin cannot be controlled by a single person or an organization because it is not centralized. The Bitcoin transactions are processed within the network with the help of a group of nodes that work together to mine the Bitcoin cryptocurrency, which does not require any form of a central authority.

4.7.2 Ethereum

Ethereum, which is based on blockchain technology, is an open-source software platform. Ethereum enables developers to build and deploy decentralized applications like smart contracts. It offers a decentralized virtual machine—Ethereum Virtual Machine—that can execute scripts using a globally distributed network of public nodes. It was launched by Vitalik Buterin in late 2013. The development for Ethereum was funded by an online public crowd sale by buying the Ethereum tokens (Ether) between July and August 2014.

The first public beta prerelease was the Olympic network, which provided users with a bounty of 25,000 Ethers for stress testing the limits of the Ethereum blockchain. Ethereum's live blockchain named Frontier was launched on July 30, 2015. The current release is named Homestead and is considered stable. It has led to various improvements such as transaction processing, gas pricing, and security. There are at least two other protocol upgrades planned, i.e., Metropolis and Serenity (PoS).

4.7.3 Hyperledger

Hyperledger is an open-source project created to advance cross-industry blockchain technologies. It is a global collaboration, hosted by The Linux Foundation, including leaders in finance, banking, internet of things (IoT), supply chains, manufacturing, and technology. Hyperledger acts as an operating system for marketplaces, data sharing networks, microcurrencies, and decentralized digital communities. It has the potential to vastly lessen the expense and complications in getting things done in the real world.

4.7.4 Multichain

Multichain was created by Gideon to make available a blockchain solution for organizations to build and develop quickly. Multichain is built upon Bitcoin blockchain but offers the functionality of permissions, streams, and assets, as well as other tools like explorers to interact with the underlying blockchain seamlessly. BankChain is using Multichain as its base.

4.7.5 EOS

EOS is an operating system for marketplaces, data sharing networks, microcurrencies, and decentralized digital communities. It has the potential to vastly lessen the expense and complications in getting things done in the real world (Kalodner et al., 2020). EOS blockchain aims to become a

decentralized operating system that can support industrial-scale decentralized applications. EOS is planning to delete transaction fees, and it claims to have the ability to conduct millions of transactions per second. EOS runs on DPOS consensus algorithm.

4.7.6 Corda

Corda is an open-source blockchain aiming to meet requirements for use of blockchain in businesses. Corda offers a solution where a shared ledger can be initiated between parties under a contract. Corda's communications are point-to-point, meaning only participants of a transaction can see it. With Corda's point-to-point architecture, participants only have copies of the transactions where they are observers or participants. A unique ledger is present in every node of Corda's network, which is called a multilateral ledger.

4.7.7 NEO

NEO is a smart economy for the distributed network (Yeoh, 2017). NEO was chosen as the name because neo in Greek means, newness, novelty, and youth. NEO was initially called AntShares (ANS), which was launched in 2014, founded by Da Hongfei and Erik Zhang. AntShares announced on June 22, 2017, that it planned to rebrand itself as NEO. The first initial coin offering (ICO) on the NEO blockchain, Red Pulse Token (RPX), was announced soon after the rebranding finished. Apart from the NEO cryptocurrency itself, the NEO platform has another crypto token, GAS, which was formerly called ANC-Ant coin.

4.7.8 Quorum

Quorum is the brainchild of J.P. Morgan, bringing in an enterprise-focused version of Ethereum. Quorum blockchain delivers private, smart contract implementation and industry-grade output. Quorum uses zk-Snarks cryptography, which allows verification of the computation correctness without even learning what was executed. Quorum uses a hybrid privacy design.

5. Evolution of blockchain in fields other than cryptocurrency

Since the introduction of blockchain technology, there has not been much progress of blockchain technology other than cryptocurrencies. But this technology can be used in various other fields such as politics, music, IoT, supply chain, etc., as shown in Fig. 1.7. Voting is the most important part of

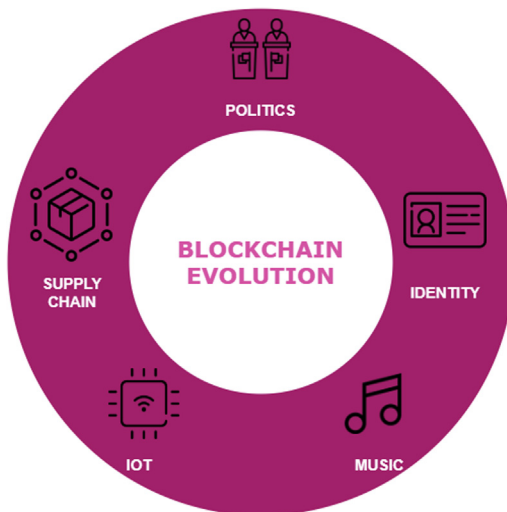


Figure 1.7 Evolution of blockchain.

a democracy. Voter fraud is a critical issue today, which can be easily resolved by using a blockchain. For an instance, a blockchain can be created for voting where each person or node is then added to the blockchain, followed by issuing a single nontransferable token to each node, so each person has a single chance to vote. The tampering of elections in the case of blockchain is impossible because of the large computation power required to change the information at all nodes since one should have complete command over 51% of the network.

In the entertainment industry the movie *Braid* was the first movie to be produced by doing an ICO (Nguyen and Kim, 2018). In property rental, Rentberry aims to address the common pitfalls and headaches of the traditional rental model. Regarding blockchain in politics, Sierra Leone carried out their elections on blockchain. In education, the Socrates Coin is making big moves to change the traditional approach to a 3D internet. In digital advertising the Basic Attention Token is taking a crack at solving advertisement problems. In the IoT sector, Walton chain is an award-winning Chinese project that seeks to integrate IoT and blockchain technology on an unprecedented scale.

6. Challenges of blockchain

As in every other technology out there, blockchain also has its share of challenges. One of the biggest roadblocks for this emerging technology is

illegal activity. Confidentiality is a key factor of blockchain as it protects the users of the network from hacks and also preserves their privacy, but unfortunately, it can also be used for illegal activity. Some other challenges that blockchain faces are high cost, inefficacy of speed, and that it cannot be maintained if the network grows too large.

6.1 Industry challenges with blockchain

6.1.1 Energy consumption

Some major public blockchains use PoW algorithms, which involve the use of the computational power of a machine to solve a complex mathematical puzzle to verify a transaction and add it to a block. Current Bitcoin energy consumption is almost equal that of Ireland. By 2020, it is estimated that Bitcoin will utilize more energy than the entire world currently uses (Kshetri and Voas, 2018). A probable solution for this has emerged in the form of consensus mechanisms like PoW and PoS.

6.1.2 Scalability

Scalability has appeared as a significant issue for blockchain networks like Bitcoin and Ethereum. Blockchains are having issues with efficiently holding a large number of nodes on the network. Moreover, the size of public blockchains keeps on increasing. Currently, Bitcoin ledger size is greater than 100 GB. One possible solution that has emerged is storing a hash of data over the network.

6.1.3 Public perception

Presently, blockchain technology is almost synonymous with Bitcoin. A majority of the public is still unaware of the presence and potential practices of blockchain technology (Chauhan et al., 2018). Since Bitcoin is anonymous, it is used for mysterious transactions of black-market trade, money laundering, and other illegitimate events, giving it a bad reputation. Mainstream adoption is needed to remove the sometimes negative undertones of Bitcoin when misused.

6.1.4 Blockchain standards and regulations

Blockchains are continuously evolving, but countries are still skeptic about them because there is no proper definition for standards and regulations. Enterprises and governments require regulations to protect their customers. To tackle this problem, certain countries are trying to launch their own regulations for the technology. Mass adoption might also standardize the blockchain.

6.2 Attacks against blockchain

6.2.1 Sybil attack

In a Sybil attack, the assailant tries to plug the network with the client's nodes that they control. If this happens, then you would be most likely to join with attacker nodes. Bitcoin never keeps a count of nodes for anything, so if the attacker completely isolates a node from the authentic network, then it can help the attacker in the implementation of other attacks. The potential damage could be that the attacker refuses to relay blocks or possibly the attacker only relays blocks that he creates.

6.2.2 Eclipse attack

This attack is based on distributed application architecture that partitions errands or assignments among users without the necessity for a central server that is coordinating or steady hosts. Crippling a node in such a way means it cannot talk to other nodes in the network. This attack is possible due to design strategy flaws in the blockchain peer's identity and peer selection strategy. Currently, Bitcoin has eight outgoing connections, and Ethereum has 13, which imply one node in Bitcoin only has a view for eight nodes connected to it. So, one node in Bitcoin has to depend on the other eight for the complete view of the network, which can be taken advantage of by a hacker ([Sankar et al., 2017](#)).

The potential damage could be double spending, attacks against second layer protocols, or an assailant acquiring services or products without paying by deceiving his victims into thinking that the channel of payment is still open while the other part of the network sees that channel of payment is closed. Smart contracts also may be compromised if peers of the blockchain network see unreliable records over the blockchain.

6.2.3 Time-jacking attack

Time-jacking attack is an extended version of the Sybil attack. In this attack, each computer involved keeps a time counter for the network. The network time counter works on the median time of a node's peers, which is sent in the version message when peers are connected. The network time counter reverts to the system time if the median time varies by more than 70 min from the time of the system ([Li et al., 2020](#)).

The potential damage could be that an assailant could possibly speed up or slow down a computer's network time counter by linking multiple nodes and broadcasting imprecise timestamps. Since the time value can be distorted by at most 70 min, the time alteration between the peers would be 140 min.

6.2.4 Fifty-one percent attack

Basically, a 51% attack takes place when a group of miners get control of more than 50% of the mining hash rate of the network, or a group of miners with substantially more computational power can get full control of the network. It is a speculative attack described over Bitcoin blockchain. Bitcoin Gold, when it was among the top cryptocurrencies, suffered a 51% attack and lost (Khan and Salah, 2018).

The assailants would be able to stop fresh transactions in advance, gaining authorizations and making them stop payments among users. Attackers would also be able to redo the transactions that were established while they were in control of the blockchain network, meaning they could double their coins. The mining pool ghash.io was momentarily surpassing 50% of the Bitcoin network computation power in July 2014, leading the pool to obligate to dropping its share of the voluntarily reduced its share of the network.

7. Applications of blockchain

The blockchain technology has several applications, as shown in Fig. 1.8, other than storing transaction data. This technology has already been incorporated by several big companies like Walmart, Pfizer, AIG, and IBM. For instance, IBM has used blockchain in the Food Trust blockchain, which helps trace food products and their location. The reason behind doing this is to check if food is suspected to be poisoned or contaminated. If they went with a traditional database to manage and store the information of the supply chain of food products, it would take forever to trace back where the contamination occurred.

But by storing this data using blockchain, then every part of the journey is recorded and stored in a block that is time stamped, and it is very easy to trace the food product route. This helps to save innumerable lives if the food contamination is massive. Besides the food industry, blockchain can potentially be used in any field. Some of the fields that are using blockchain are banking and finance, currency, healthcare, property records, and state identification (Zhang et al., 2019).

7.1 Mortgages and loans using blockchain

The problem with the current day-to-day loan and mortgage system is that it has a lot of mediators and third parties are involved, typically banks. The process of getting a loan from a bank is very time consuming and

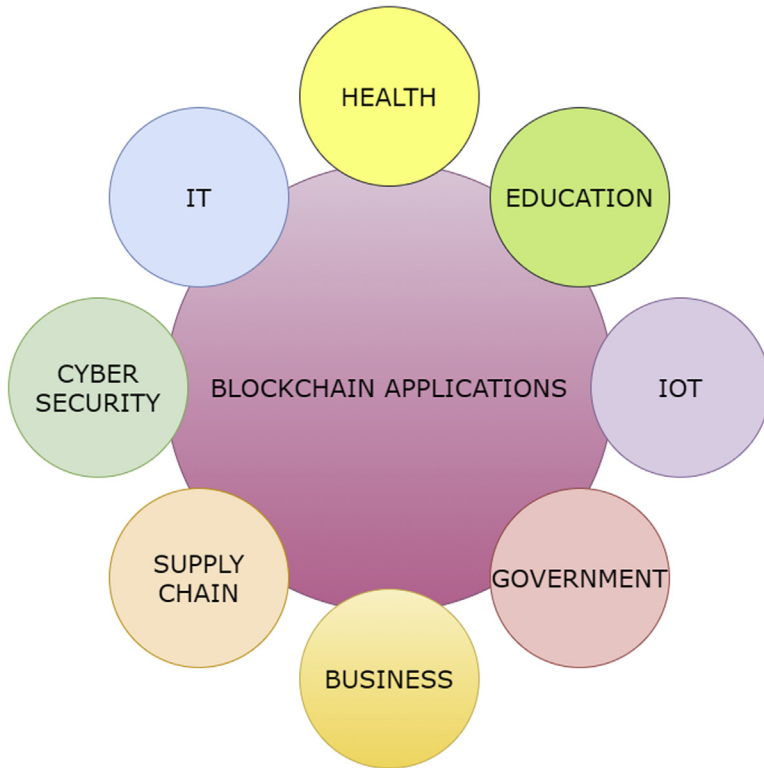


Figure 1.8 Applications of blockchain.

complex and also involves a ton of paperwork. A blockchain-based platform with a network of mortgage and loan lenders seems to be the best course of action for the problem that we are facing with the mortgage and loan system.

This process of using a blockchain requires the lenders and borrowers in the blockchain network to form smart contracts, which then issue tokens for each mortgage or loan issuance. As soon there is a deadline the blockchain automatically implements the details present in the smart contract uploaded by the borrower and lender. The main benefit from using this type of mechanism for mortgage and loan issuance is that it is very cost affective and saves a ton of money and time dealing an intermediary. The use of a blockchain in this scenario increases the transaction performance, and the speed of each transaction can be increased (Park and Park, 2017). Usually in the case of loans, there are many cases going on in the civil courts, but with the use of a blockchain, the transparency for borrowers and

lenders is increased. This proposed method of using a blockchain to process loans and mortgages is much more efficient and straightforward, reducing the processing time.

7.2 Cross-border payments using blockchain

Generally, the problem with the current system of cross-border payments is they cost a lot of money and consume a great amount of time. This process is opaque and the time for each transaction is pretty high. The solution to this problem would be the use of a blockchain with smart contracts and digital assets. The smart contracts take care of the users involved in the financial agreements. The digital assets help to reduce the cost by cutting the capital compliance and operational cost, which helps the user to save money compared to the traditional way of payments involving cross-border payments.

Different types of currencies can be used for transactions for cross-border payments, which typically involve banks. The benefits of using this system are by using a blockchain the transaction becomes faster than a typical transaction. Because of the use of blockchain, there is no involvement of a third party, which helps the user to save a lot of money and also reduces the risk of tampering with the transactions. More security and transparency can be achieved by shifting to blockchain technology. Another alternative to this cross-border payment is the use of digital cryptocurrencies as they solve the liquidity issues of cross-border payments (Karame and Capkun, 2018).

7.3 Stock trading using blockchain

The problem with stock trading using existing technology is that the functioning of stock exchanges includes dealings that consume a lot of time and are very complex. Stock trading is susceptible to risks and is cost inefficient. By using a private blockchain, one can build a trading platform that allows the companies to register and sell their shares digitally over a public blockchain. The investor can get complete transactions benefits and records securely and with ease.

With the use of smart contracts, the stock's movement is recorded, and after one agree to the terms of logical transaction, the money will be deducted from one's account, and the blockchain validates the profit or loss using smart contracts. After validating, the amount would be debited or credited to their account. The security properties settlements can be given

by issuing digital tokens. The elimination of intermediaries or any third parties is the main benefit of this methodology of using a private blockchain. There are a couple of benefits of using a private blockchain for stock trading; one of them is that the stock can be traced back to the buyer, and another is that settlements can be done much faster using this technology.

7.4 Online identity management using blockchain

The problem with the existing technologies of online identity management is that it is a time-consuming and costly process. A permission-based blockchain to store digital identities would be the solution to tackle this real-world problem of managing online identities. Based on the immutable nature of a blockchain the user identities cannot be tampered with. The users in this permission-based blockchain have the liberty to choose how one can see them in the blockchain and with whom the personal information can be shared.

This blockchain can use the digital identity in various ways, for example, to apply for loans or mortgages. This helps to maintain a single and ultimate source of identity (Halpin and Piekarska, 2017). The benefits of using this permission-based blockchain are that any intermediary can be avoided and every user can be provided with same level of security. Information can be accessed anywhere and anytime, as it is a single source of identity. Unlike the traditional way of authenticating identity, there would be no reason for authenticating multiple times, as this is an immutable, legitimate manner of verification.

7.5 Public health security using blockchain

Health is the foremost issue that should be taken into consideration by governments. But the technology that we use to store and maintain public health records is primitive and has many loopholes that can be exploited. The main problems are with authenticating and maintaining public health records. With the use of a public blockchain in which the users are patients, hospitals, and healthcare providers, this problem of storing, maintaining, and authenticating health records can be resolved.

The health records of all patients should be stored and maintained using a public ledger distributed system, a blockchain. After creating a patient record, a key should be set up by the user so one has access to the user's personal health information. The initialization of smart contracts between the hospital and the user is a very good use case application. An IoT device

can be used to transfer the data of the patient directly over to the blockchain, so at a later date, the concerned healthcare provider has access to all the patient's health information (Halpin and Piekarska, 2017).

The main benefit of using this blockchain to store and maintain public health records is to have complete data security for the users of this blockchain network. With only the proper permissions can one access or give access of their personal health information to whomever they require, typically healthcare providers and pharmacies. The use of a public blockchain also reduces the complexity of the data sharing agreements compared with traditional public health security.

7.6 Tracing of medicine using blockchain

The problem with traceability of medicine is that there are a lot of counterfeit drugs present on the market that cause a potential danger to people; some of them can even be fatal. With the use of a blockchain for the medicine supply the problem of traceability of medicine could be resolved. A unique identification should be issued to each medicine to help trace every step of the supply chain. For suppliers that have approval of the government a different ledger can be formed, and the documents related to the reliability of the medicine should be stored and maintained in the blockchain for verification and authentication.

In this scenario the use of smart contracts would be appropriate between the distributors and suppliers. The benefits of using a blockchain would when medicine is counterfeit or not based on the data available on the blockchain. Cost can be reduced once counterfeits go out of business. Having fewer counterfeits helps to protect people using medicines as well as helping to preserve the reputations of pharmaceutical companies.

7.7 Rental property using blockchain

The problem with the rental properties is that multiple listing services provide lists that are incorrect and incomplete. Also, the ownership and documents of the rental properties are not well defined. Using a public blockchain for storing and maintaining rental property would be a very good solution, and by using the blockchain the payments of rents and related services would become much easier for all concerned parties.

The use of a public blockchain for storing and maintaining rental records enables records to be stored in one place, and all the transactions and listings could be authenticated independently, and they could be automatically submitted by the use of smart contracts. The rental process would be executed with the support of self-implementing smart contracts over the

blockchain. Since the transactions between the tenant and landlord can be automatically done by using smart contracts, there is no need for third-party real estate agents or any other sort of intermediary.

The use of blockchain greatly reduces the risk of alteration of rental records. All this information is stored in every computer present in this network, so in case of any attempt at tampering or altering of data, the data can be retrieved from the blockchain network. The usage of this proposed system reduces trust issues and other miscommunication involved with rental records, saving time and money for both the landlord and tenant.

7.8 Property records using blockchain

The problem with property records is that they are stored and maintained at the local government level. If there is tampering of property records on the government side due to corruption or some natural disaster, there is chance that we can lose our properties. Having centralized control over property records is a sensitive trust issue. Using a multiledger distribution blockchain, where each sector of government stores and maintains the land records per their authority (Dasgupta et al., 2019), allows for a secure environment. These records are accessible to every person whose property records are present in this multiledger blockchain, if they provide correct identification, and there will be no loss of information or corruption of property records since they are stored in a blockchain. The usage of tokens can be encouraged to perform transactions between two parties, like the transfer of ownership of the property records.

Smart contracts can also be used to upload the contract statement, and the blockchain withholds the cryptocurrency or property records and transfers them to the concerned party after validating the contracts of both parties. A benefit of using a multiledger distributed blockchain in the case of property records is that it reduces the trust issues between a government and its people. The use of a multiledger blockchain drastically reduces the risk of tampering with property records. This type of system increases the chances of a middle-class person to invest in small properties. As this is a blockchain, all this information is stored in every computer present in this network, so in case of scandal or natural disaster the data is not lost and can be retrieved from other nodes present in the blockchain network (Karuppiah and Saravanan, 2014, 2015; Karuppiah et al., 2016, 2017, 2019; Kumari et al., 2016; Naeem et al., 2020; Maria et al., 2021; Pradhan et al., 2018; Li et al., 2017).

7.9 Real estate regulations using blockchain

The problems with regulations regarding real estate is that they are not transparent and not being used effectively. The main purpose should be transparency, and there should be universal accessibility to users involved in a real estate contract. But that is not the case in current times. This problem can be resolved by the use of a public blockchain in which the real estate regulations are listed, so they cannot be tampered with, making them accessible to users involved in a contract.

If real estate regulations are listed using a public blockchain, they would be transparent to every party in the value chain. The regulations can be arranged on a public, distributed ledger as a transaction and posted on the public blockchain. Transactions can be done with ease as there is transparency in associated timelines and costs. Using a public blockchain is superior to traditional processes, where even the government profits from the ease and transparency.

7.10 Liquidating real estate assets using blockchain

The problem with liquidation of real estate is that it takes many days to complete a transaction using existing technology. In the current environment, liquidating real estate assets is a hectic process. This problem of taking so much time for a single transaction can be tackled by creating a public or private blockchain that increases the speed of each transaction, as well as being transparent and efficient (Dasgupta et al., 2019).

To sell or buy property over the blockchain, one can be issued tokens to perform transactions. Since these transactions take place using a blockchain, there is less chance for disputes over the ownership of a property. All the transactions that take place in a real estate blockchain can be accessed by anyone in the network and ultimately are transparent. Even financiers and money lenders can get a maximum reach through this blockchain and a new approach to the liquidation of real estate assets. There are also benefits over the traditional process in that there is no third party involved in the transactions, so the transfer of ownership is done within minutes, which saves a lot of time and money.

8. Conclusion and future scope

Blockchain technology is not limited to cryptocurrencies like Bitcoin. It has a vast array of applications in sectors such as government, medicine, energy,

etc. Blockchain technology provides decentralization, efficiency, security, and privacy for its applications. In the future, we can see the usage of blockchain technology in almost every sector, which can transform the world to a much safer place. Blockchain removes the trust gap that we have seen for generations. Before, we used intermediaries like banks to do financial transactions, which are not very safe for a variety of reasons such as centralization, lack of transparency, etc. But with the use of cryptocurrencies, there is no need to trust an intermediary, and we can overcome almost all the drawbacks caused by intermediaries.

Blockchain is a rising innovation. Consequently few designers are worried about the standards of computer programming applied to blockchain-based frameworks. By the by, the number and assortment of blockchain executions keeps on expanding. Also, the absence of rules and guidelines on the most proficient method to plan programming structures that incorporate keen agreements as a feature of the framework calls for additional attention. Therefore, adopters should zero in on choosing the arrangement that best meets their requirements and the necessities of their decentralized applications as opposed to growing one more blockchain without any preparation. Using this blockchain technology, there is a less chance to falsify property information, due to the transparency of blockchain, so real estate investment trusts can also be operated by the usage of issuing tokens for properties.

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